PRE-DATATHON WORKSHOP

NU DATA SCIENCE CLUB

HOW TO WIN THE DATATHON



- 19:00
- 16.10.2026
- Room 7E.329

DATATHON '26

3,000,000 ₸

PRIZE POOL

ORGANIZER

DataSci

CASE PARTNER

ForteBank

SUPPORTED BY

NU Impact Foundation



WHO WE ARE · SPEAKER 1

FATIKH BARYSH AIDYN

AI/ML engineer • CS at Nazarbayev University

ForteBank	AI Engineer	SINCE SEP 2026
ArqaMed	Software Engineer	SINCE AUG 2026
Halyk Bank	AI Engineer	MAY-AUG 2026
ISSAI NU	ML Intern	FEB-MAY 2026
nFactorial	Full Stack, Incubator	JUN-AUG 2025

- Today: the format and strategy



WHO WE ARE · SPEAKER 2

DAUREN APAS

Software engineer and mentor • CS at Nazarbayev University

ESEP	Software Engineer	SINCE OCT 2025
ArqaMed	Software Engineer	SINCE AUG 2026
nFactorial	Tech Mentor	SINCE JUN 2026
nFactorial	AI Engineer, Incubator	MAY-AUG 2025

- Today: standing out and the pitch

TEAM RICHARDS

OUR WINS

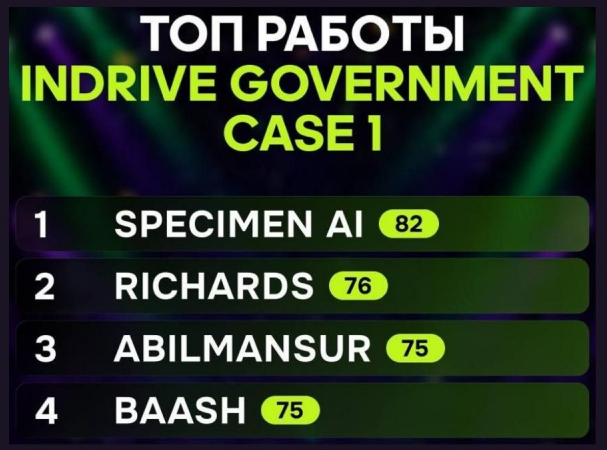


1ST PLACE

HackNU 2025

Higgsfield ML Track

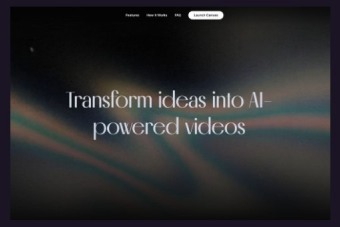
300 000 ₸



2ND PLACE

Decentrathon 5.0

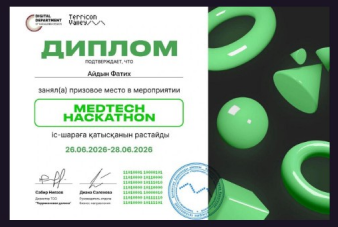
inDrive Government Case 1: Legislative Entropy



2ND OF 180

De-Vibed

300 000 ₸



3RD PLACE

MedTech Hackathon

100 000 ₸



1ST PLACE

GenerativeX 2026

125 000 ₸

BEFORE WE START

THE BEST SOLUTION DOESN'T WIN.

Hands up if you agree.

HOW YOU PRESENT IT MATTERS AS MUCH AS WHAT YOU BUILT.

The jury only knows what you show them.

PRESENTING IS A SEPARATE SKILL.

Being good at building doesn’t make you good at explaining.
Practice it too.

LESSON 1

JUDGES CHECK THE LIST.

Hour 1: turn the brief into a checklist. In the pitch, tick every box on screen.

LESSON 1 · WHAT IT MEANS

COMPLETE BEATS IMPRESSIVE.

TEAM A

- ✓ Strong model
- ✓ Fancy dashboard
- ✓ A chatbot nobody asked for
- ✗ Explanations
- ✗ README

Impressive. Incomplete.

TEAM B

- ✓ Solid model
- ✓ Explanations
- ✓ Working demo
- ✓ README
- ✓ Every item in the brief

Simple. Complete. Wins.

QUESTION

YOUR MVP TICKS EVERY BOX.

8 hours left. What do you do?

THE TRAP

A

Add a cool new feature

B

Make what you have better

C

Work on the pitch

LESSON 2

GO DEEPER, NOT WIDER.

- Harden the demo
- Measure it
- Sharpen the explanations
- Record a backup video
- Finish the README
- Rehearse

LESSON 2 · WHAT IT MEANS

FINISHED BEATS MANY.

WIDER

Search

Dashboard

Chatbot

Login

Export

Lots of features. None finished.

DEEPER

Search

Results

Explanations

Few features. All of them work.

LESSON 3

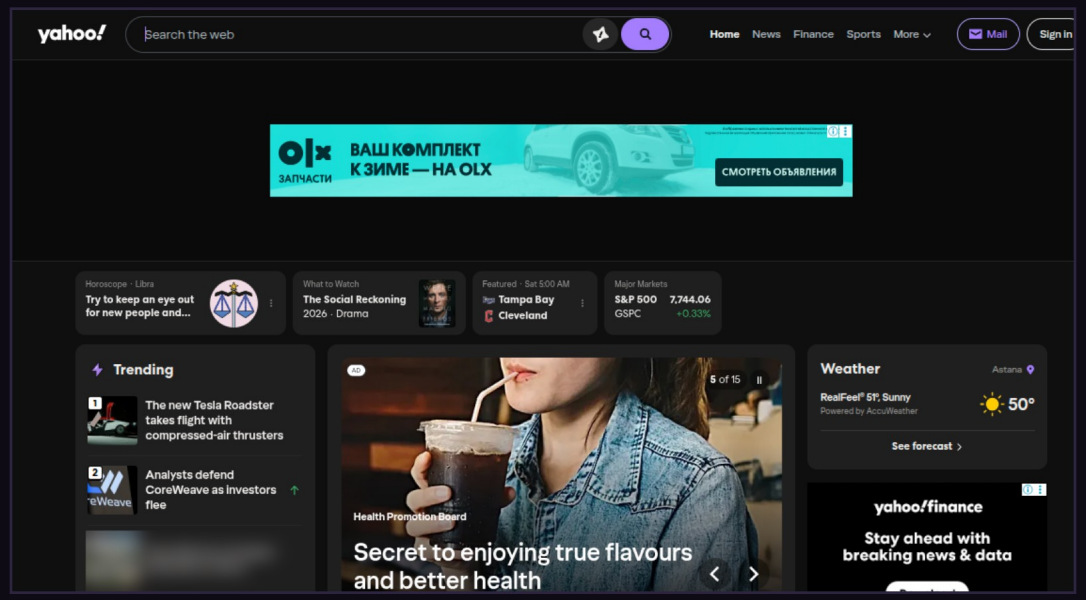
SIMPLE BEATS FULL.

One clear flow. Nothing the user didn’t ask for.

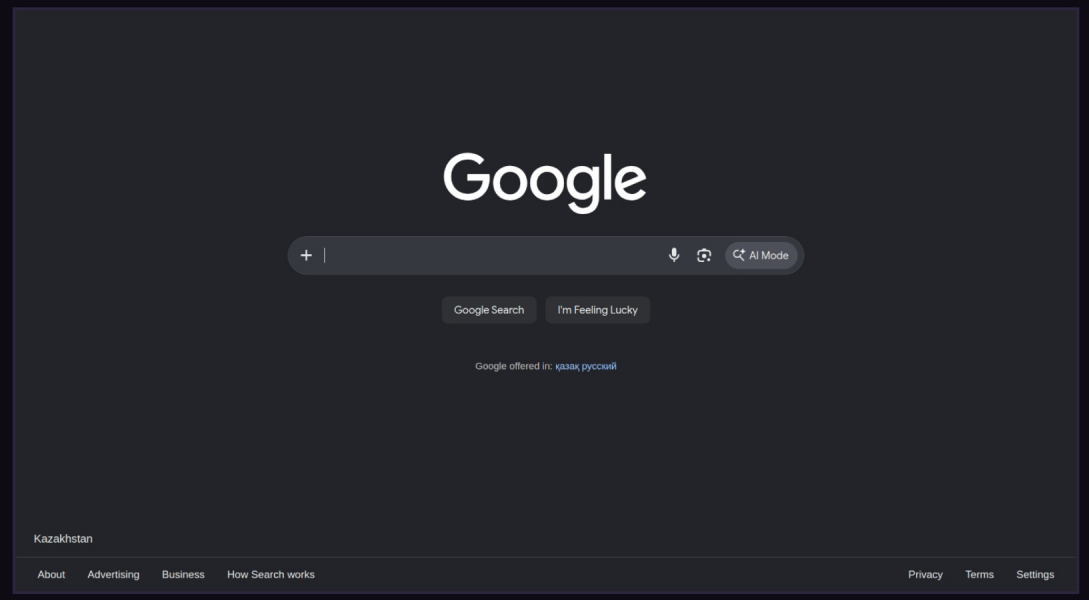
LESSON 3 · WHAT IT MEANS

SAME JOB. DIFFERENT INTERFACE.

Same job: search. One is cluttered, one is simple.



TOO MUCH



JUST ENOUGH

LESSON 4

SHOW THE WORK, NOT THE STACK.

Your slides are for what you built, not what you built it with.

SAME PROJECT. TWO SLIDES.

One shows the tools. One shows the work.

Our Tech Stack

React	FastAPI	PostgreSQL
Docker	LangChain	OpenAI API

THE STACK

What our system found

Screenshot of the product working

-40%
time to review

Why it works: one sentence, in plain words

THE WORK

THE EXCEPTION

NICHE TECHNIQUE? NAME IT. CITE IT.

A “We trained a CNN and got 94% accuracy.”

B “The CNN was cheating on background cues, so we retrained it with **Group DRO** (Sagawa et al., ICLR 2020).”

WORST-GROUP ACCURACY: [__%] → [__%]

Name the technique • cite the paper • show what it fixed

BUILD SMART · 01

SET UP WHAT YOU CAN. TONIGHT.

Pitch deck template
Problem, solution, results, team: empty slides ready
→ Tomorrow you just fill it in. No hours on design.

Environment
Python, libraries, Jupyter, API keys
→ No pip install while the clock runs.

Code skeleton IF ALLOWED
Data loading, quick EDA plots, baseline, eval
→ You start on the case in minute one.

Front-end and deploy IF NEEDED
Datathon = data first. UI is a bonus.
→ A demo without building UI from zero.

BUILD SMART · 02

ANALYZE THE JURY, NOT JUST THE CASE.

THE CASE

Clear as it is.

Every team reads the same one.

THE JURY

- Who are they?
- What do they expect from you?
- What do they need?

Every jury has different expectations and different needs.

BUILD SMART · 03

PICK A PATH. ADAPT ON THE WAY.

TOO MANY IDEAS?

Try one idea

↓

Check it after a few hours

↓

Not good? Switch, with time left

HOW WE USUALLY DO IT

Choose one path as a team

↓

Build it

↓

Add, swap and fix parts in place

BUILD SMART · 04

DIFFERENT ISN'T BETTER.

STANDING OUT WITH THE TECH

An exotic model

Custom-built from scratch

A brand-new framework

STANDING OUT WITH THE RESULT

An insight others missed

A number that proves it

A demo that just works

LAST DATATHON We built our own model to look different. Others used an GEMINI and beat us.

BUILD SMART · OS

DEPLOY IS A BONUS.

2 **Then deploy**
A live link puts you ahead

1 **The best solution you can make**
A deployed bad solution is still bad

A last-minute deploy error eats your pitch time.

LOW ON TIME?

Skip the deploy.

Prep the pitch and harden the demo instead.

SHOW WHAT WORKS.

ON STAGE

Data in → **Model runs** → **Result + why**

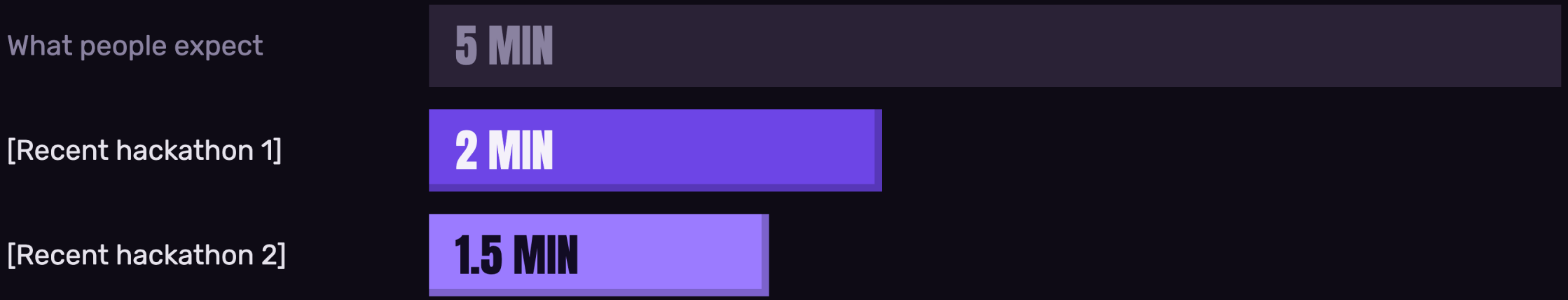
EXISTS, BUT NOT ON STAGE

- Edge cases
- Slow parts
- Half-done features

Asked about it? Answer honestly. Don’t bring it up yourself.

BUILD SMART · 07

YOUR PITCH MIGHT BE 2 MINUTES.



Prep early. Train with a timer.

⟨WE DIDN'T. DON'T BE US.⟩

QUICK ONES

FOUR SMALL THINGS

01 Check your result the way the jury will.

02 Don’t explain the problem. The jury already knows it.

03 Plan one moment they’ll remember.

04 Use tools you already know.

BACK TO THE START

THE BEST MODEL DOESN'T WIN.

So what does?

01

Every requirement covered

02

Depth, not extra features

03

A simple interface

04

Slides that show the work

TOMORROW · 11:00

ONE DATASET.
ONE PROBLEM.

That’s all we can tell you. We don’t know more either.

WRAP-UP

TONIGHT'S CHECKLIST

1

Laptop, charger, extension cord

2

Check-in at 09:00, Orange Hall

3

Telegram channel: @nu_datasci

4

Sleep. Seriously.

FAQ

QUESTIONS?

Ask us anything about tomorrow.

WEBSITE	TELEGRAM	EMAIL
datathon.com.kz	t.me/nu_datasci	datascience@nu.edu.kz

SEE YOU TOMORROW · 09:00 · ORANGE HALL

